

PAPER_1078: QCalcGeom Master Equation Derivation

Star Magic UQFF Framework — Session 225

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Abstract

We derive the QCalcGeom master equation combining Breit-frame Schwarzschild-flux-graviton (BSFG) crossover radius, 26-dimensional compactification scale, Ramanujan polylogarithmic corrections, and SCm phonon fluence into a single length-scale observable:

$$\text{QCalcGeom}(r, \Gamma) = r_{\text{cross}} \cdot (26!)^{-1/13} \cdot S_{26}^{(3)}([SSq]) \cdot \Phi(\omega, \Gamma)$$

1. BSFG Crossover Radius

The crossover radius marks the transition between quantum-gravitational and classical regimes:

$$r_{\text{cross}} = \sqrt{\eta_{\text{BSFG}}} \cdot \frac{GM}{c^2}$$

where $\eta_{\text{BSFG}} = 10^{-22}$ is the BSFG coupling and GM/c^2 is the gravitational radius. For the Sun ($M = 1.989 \times 10^{30}$ kg):

$$r_{\text{cross}} = \sqrt{10^{-22}} \cdot \frac{6.674 \times 10^{-11} \times 1.989 \times 10^{30}}{(2.998 \times 10^8)^2} = 1.477 \times 10^{-8} \text{ m}$$

2. 26D Compactification Scale

The compactification scale arises from the 26-dimensional string compactification volume:

$$\ell_c = (26!)^{-1/13} = (4.033 \times 10^{26})^{-1/13} = 8.983 \times 10^{-3}$$

This sets the dimensionless ratio between compact and extended dimensions. The first five Kaluza-Klein eigenvalues are $\lambda_n = n(n + 25)$ for $n = 1, \dots, 5$: {26, 54, 84, 116, 150}.

3. Ramanujan Polylogarithmic Sum

The $S_{26}^{(3)}$ correction factor:

$$S_{26}^{(3)}([SSq]) = \sum_{n=1}^{26} \frac{[SSq]^n}{n^{26}} \cdot R_n^{(3)}$$

where $R_n^{(3)}$ is the Ramanujan correction factor. With $[SSq] = 0.57$:

$$S_{26}^{(3)} = 9.500 \times 10^{-2}$$

4. Phonon Fluence Factor

$$\Phi(\omega, \Gamma) = \exp\left(-\frac{(\Gamma - \Gamma_0)^2}{2\sigma_G^2}\right) \cdot S_{26}^{(3)}$$

This Gaussian envelope peaks at $\Gamma_0 = 2\pi \times 0.10$ THz with width $\sigma_G = 0.08 \times 2\pi \times 10^{12}$ rad/s.

5. Master Equation Assembly

Combining all four factors:

$$\begin{aligned} \text{QCalcGeom}(M_\odot, \Gamma_0) &= 1.477 \times 10^{-8} \times 8.983 \times 10^{-3} \times 9.500 \times 10^{-2} \times 9.500 \times 10^{-2} \\ &= 1.197 \times 10^{-12} \text{ m} \end{aligned}$$

6. Dimensional Proof

The SI unit trace confirms correct dimensionality:

- r_{cross} : [m] (length)
- $(26!)^{-1/13}$: dimensionless
- $S_{26}^{(3)}$: dimensionless
- Φ : dimensionless
- Product: [m] – a **length scale**

7. Validation

- Linewidth sweep shows peak at $\Gamma = 0.10$ THz
- Linear mass scaling: $\text{QCalcGeom}(2M) / \text{QCalcGeom}(M) = 2.0000$
- All 10 self-tests pass

References

1. BSFG coupling: `qcalcgeom_helpers.py` ($\eta = 10^{-22}$)
2. 26D string compactification: `MAIN_1_CoAnQi.cpp` SOURCE115-116
3. Ramanujan corrections: `vds_dvp_bsh_symbolic_proofs.py`
4. SCm phonon framework: `scm_phonon_linewidth.py`